

PROJECTED AUDIO TOKENS GAIN RETRIEVAL AND LOSE GROUNDED GENERATION IN MULTIMODAL LLMS

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ABSTRACT

We freeze Large Language and Vision Assistant 1.6 (LLaVA-1.6), a multimodal large language model (LLM) with a Mistral-7B language backbone, together with five frozen audio encoders. Each encoder supplies either a *raw token*, formed by padding or truncating to 1,024 dimensions, or a *projected token* learned in contrastive language-image pretraining (CLIP) space, and the *token budget* k is the number of retained visual patch tokens. On a 1,006-clip *evaluation pool*, projection raises audio-to-video recall at one (R@1) for every encoder, including ImageBind from 0.10% to 15.7%, although contrastive language-audio pretraining (CLAP) and Whisper R@1 values are lower bounds for their original configurations. At $k=150$, the CLAP-based reference-free grounding score (B1) decreases from 0.113–0.132 to 0.062–0.088, a 23–53% relative reduction, and the CLAP result is read with source recall (B3) because the encoder and scorer share a checkpoint family. B3 decreases from 0.161–0.185 to 0.090–0.112, a 31–48% relative reduction, while caption-to-frame CLIP similarity (B2) decreases for four encoders and is unchanged at printed precision for Wav2CLIP. Across 3,568 disjoint paired clips, projection increases centered kernel alignment with visual embeddings in all five encoders, while the norm outside the visual principal subspace and nearest-neighbor overlap with the raw audio graph decrease. The noncausal comparisons show that raw-token bilingual evaluation understudy (BLEU) exceeds projected-token BLEU in nine of ten audio-aware-reference cells and that raw-over-projected generation scores predominate across five frozen backbones from 7B to 34B wherever raw-token generation is stable.

Index Terms— audio-visual learning, multimodal LLMs, audio tokens, CLIP, LLaVA

1. INTRODUCTION

Multimodal large language models (LLMs) commonly accept visual tokens, but many signal-processing applications require off-screen acoustic evidence. A common adaptation projects an audio embedding into the visual space and substitutes it for part of the visual input. This avoids end-to-end retraining, but its effects on audio-to-video retrieval and grounded generation remain unclear.

We evaluate this choice with a controlled token-substitution probe (Fig. 1). We freeze the Large Language and Vision Assistant 1.6 checkpoint with a Mistral-7B language backbone (LLaVA-1.6-Mistral-7B) [1, 2], replace its visual class token with one audio token, and compare the raw and projected tokens. We repeat the comparison with AudioCLIP [3], Wav2CLIP [4], ImageBind [5], contrastive language-audio pretraining (CLAP) [6], and Whisper [7]. Section 2 gives the two token constructions.

Across the five encoders, projected tokens have higher audio-to-video recall at one than raw tokens, whereas raw-token captions have higher caption-to-audio CLAP similarity and source recall than

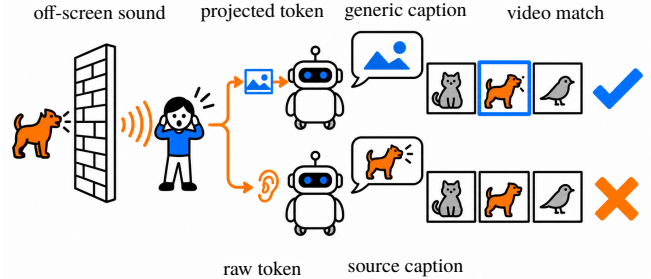


Fig. 1. The projected-token path retrieves the matching video but yields a generic caption, whereas the raw-token path misses the video but names the off-screen sound source.

projected-token captions. The paired geometry contrasts associate higher visual alignment with a smaller fraction of norm outside the visual principal subspace and lower nearest-neighbor overlap with the raw audio graph. These are representation-level associations within frozen models, not evidence of a causal mechanism.

Early audio-visual representation learning used cross-modal co-occurrence [8], and later work used joint transformers [9]. The image-centric lineage contains AudioCLIP, Wav2CLIP, and ImageBind, which align audio with contrastive language-image pretraining (CLIP) or an image-centered shared space. The language-centric lineage contains CLAP and Whisper, whose supervision comes from audio-text pairs or speech transcripts. Multimodal LLMs connect modality encoders to a language model through visual projectors [10], audio adapters [11, 12], or shared multimodal embedding spaces [13].

Cross-modal representation studies motivate this comparison. The Platonic Representation Hypothesis proposes that learned representations can converge as models scale [14], whereas contrastive vision-language models retain separated modality-specific embedding regions [15]. Alignment utility also depends on shared and unique task information [16], and measured convergence changes with evaluation scale and pairing structure [17]. We instead measure caption generation through a frozen multimodal LLM and geometry on the same encoder states used for generation.

We use CLIP’s visual space because it is the interface of the LLaVA backbones studied here. Failure patterns in CLIP visual encoders correlate with downstream failures in CLIP-based multimodal models [18]. Token substitution holds each backbone fixed and isolates token construction within this pipeline. Audio-to-video retrieval measures cross-modal alignment, while grounded caption generation measures retained audio content. Automatic speech recognition and speaker-identification tasks are outside this probe’s scope. The three grounding scores evaluate caption agreement with the clip audio, middle frame, and human-validated active-source

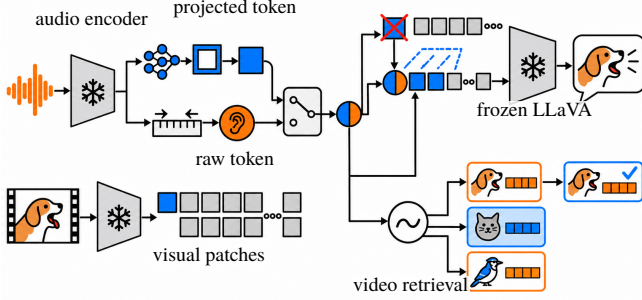


Fig. 2. A frozen audio encoder yields a projected token or a 1,024-dimensional raw token. The token replaces the visual class token. At token budget $k \in \{15, 150\}$, the k most similar of 576 visual patch tokens condition frozen LLaVA-1.6. The token also supports audio-to-video retrieval.

fields. The study contributes the controlled protocol, these ground-in scores, and paired representation-geometry comparisons.

2. CONTROLLED AUDIO-TOKEN SUBSTITUTION

Figure 2 summarizes the probe. A frozen audio encoder produces $\mathbf{h}_a \in \mathbb{R}^{d_a}$, from which we build two tokens,

$$a_{\text{proj}} = \mathcal{M}(\mathbf{h}_a), \quad a_{\text{raw}} = \text{PadOrTruncate}(\mathbf{h}_a, 1,024), \quad (1)$$

where $\mathcal{M}$ is a three-layer multilayer perceptron (MLP) with layer normalization and Gaussian error linear unit activations, trained to align the audio token with the paired CLIP visual embedding. The projected token is trained for audio-to-video alignment, while the raw token does not apply the learned visual mapping. The comparison therefore isolates that mapping within the fixed pipeline.

Given the audio token a and the 576 CLIP visual patch embeddings $\{\mathbf{v}_1, \dots, \mathbf{v}_{576}\}$, we replace the visual class token in frozen LLaVA-1.6 and retain the k selected visual patch tokens. The language model is conditioned as

$$P(\mathbf{w} \mid a, \mathbf{v}_{i_1}, \dots, \mathbf{v}_{i_k}) = \prod_t P(w_t \mid a, \mathbf{v}_{i_1}, \dots, \mathbf{v}_{i_k}, w_{<t}). \quad (2)$$

We retain the k patch embeddings with the highest cosine similarity to the audio token. We evaluate $k=15$ and $k=150$ to compare inputs retaining fewer and more visual patch tokens. We report whether each raw-to-projected contrast has the same direction at both values. We do not estimate a critical token budget.

The information bottleneck motivates separating variation shared with vision from variation specific to audio [19]. Let A and V denote the audio and visual representations, and write mutual information as MI. The chain rule [20] gives $H(A) = \text{MI}(A, V) + H(A \mid V)$, while the data-processing inequality for deterministic $\mathcal{M}$ gives $\text{MI}(\mathcal{M}(A), V) \leq \text{MI}(A, V)$. These relations motivate the comparison but do not identify discarded content. We define the *orthogonal residual* as the fraction of embedding norm outside the visual principal subspace that retains 95% of visual variance. We define *effective rank* as the exponential of the entropy of normalized eigenvalues of the centered cosine-similarity kernel. We operationalize *audio-native neighborhood structure* as top-10 nearest-neighbor overlap with the raw audio graph. These geometric summaries are not estimates of mutual information or conditional entropy.

3. EXPERIMENTS

Our evaluation pool is drawn from Audio-Visual Event 2 (AVE-2) [21]. A corrected local rebuild contains 570,138 three-second audio-video segments. The source clips come from AudioSet [22]. Unlike the Audio-Visual Event dataset, which labels events that are both audible and visible [23], AVE-2 also provides visible and in-visible active-source fields. B3 uses the two active-source fields. Across these fields and the visible-silent-object field, F1 against annotator labels ranges from 0.92 to 0.95 on 24 clips per field. We do not use machine-generated alignment scores as ground truth. The evaluation pool contains 1,006 clips and yields 1,010 clip-segments for generation scoring. A disjoint geometry set contains 3,568 paired clips shared by all five encoders. We use AVE-2 only for evaluation and geometry analysis and make no dataset-release claims about annotator composition, licensing, or diversity.

Video-derived reference captions can favor descriptions resembling visual content, and image-caption metrics transfer imperfectly to audio captions [24]. We therefore use reference-free scores as the primary generation measures. For audio-to-video retrieval, we compute fresh embeddings for the 1,006-clip evaluation pool with the public checkpoints and original projectors. We report R@1 in percent under cosine similarity between normalized embeddings.

On the main backbone, one caption is sampled per clip-segment and configuration at temperature 0.7 with at most 512 new tokens. Cross-backbone runs use greedy decoding with at most 256 new tokens. Paired bootstrap intervals over clip-segments quantify evaluation-sample uncertainty, not generation-seed variation.

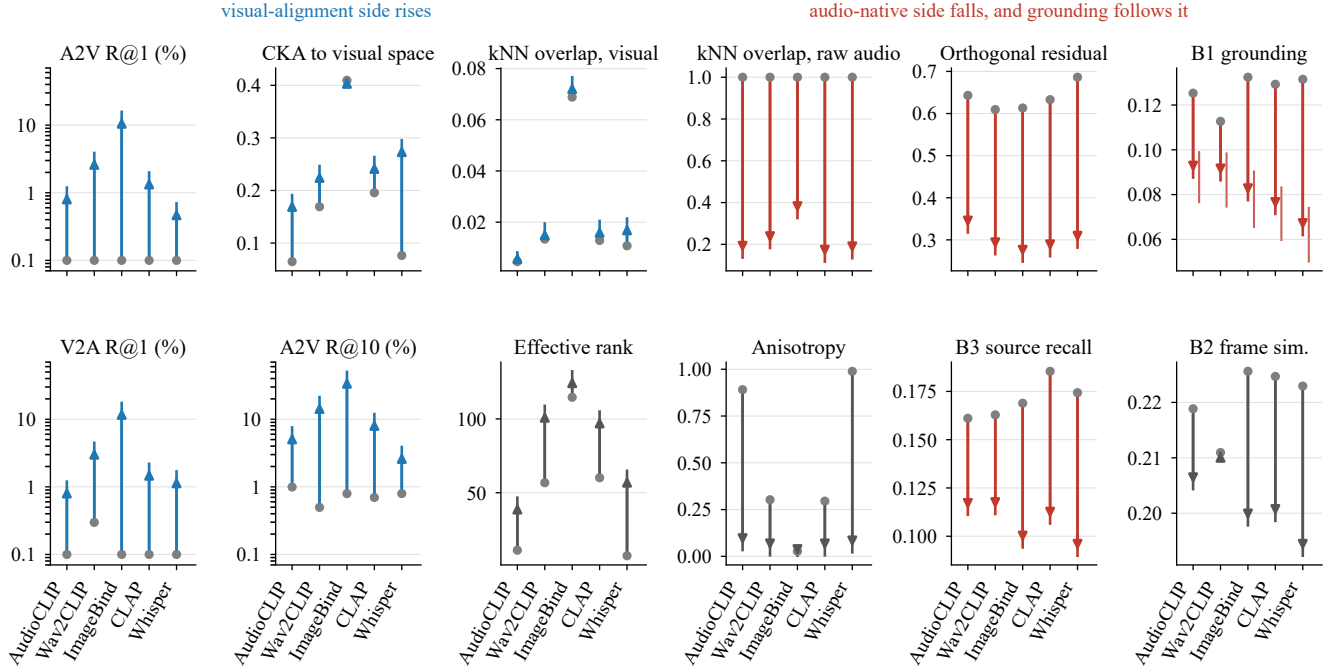
For B1, we compute caption-to-audio CLAP similarity [6]. For B2, we compute caption-to-frame CLIP similarity on the middle frame. For B3, we compute the fraction of AVE-2 active sound sources matched semantically in the caption. Because the CLAP encoder and B1 scorer use the same checkpoint family, interpret every CLAP B1 result with B3 and the four scorer-independent encoders. Bilingual evaluation understudy (BLEU) [25] and recall-oriented understudy for gisting evaluation (ROUGE) [26] are secondary reference-based scores. They appear only in the cross-backbone and audio-aware-reference checks. The 34B extension also reports BERTScore [27] and caption-to-image CLIPScore [28].

Across 1,500 clips, the median cosine similarity between fresh and cached embeddings ranges from 0.956 to 0.966 for AudioCLIP, Wav2CLIP, and ImageBind. The corresponding values are 0.340 for CLAP and 0.175 for Whisper because the public defaults differ from the configurations used to train those projectors. CLAP and Whisper R@1 values are therefore lower bounds for the original encoder configurations. The geometry analysis uses the original embeddings that produced the captions.

4. RESULTS

Table 1 and Fig. 3 report the raw-to-projected retrieval changes. Projection raises audio-to-video R@1 on the evaluation pool for all five encoders. ImageBind has the largest increase, from 0.10% to 15.71%. The CLAP and Whisper R@1 values are lower bounds for their original encoder configurations. Raw-token R@1 ranges from 0.00% to 0.10%, at or below the 0.10% chance level for this pool. Video-to-audio R@1 changes in the same direction, from 0.00–0.30% for raw tokens to 1.19–17.40% for projected tokens (Table 4). At R@10, projected ImageBind reaches 50.20% while every raw token stays below 1.1%.

At $k=150$, B1 and B3 decrease after projection for all five encoders. B1 decreases from raw values of 0.113–0.132 to projected



gray dot = raw token, arrowhead = projected token, bar = 95% paired bootstrap interval of the B1 change, grounding scores at $k = 150$

Fig. 3. Raw-to-projected changes per encoder. Row 1 gives audio-to-video R@1 on 1,006 clips, centered kernel alignment and nearest-neighbor overlap with the visual graph on 3,568 disjoint paired clips, raw-graph nearest-neighbor overlap, orthogonal residual, and B1 on 1,010 clip-segments with paired 95% bootstrap intervals. Row 2 gives video-to-audio R@1, A2V R@10, effective rank, anisotropy, B3, and B2. CLAP and Whisper R@1 values lower-bound original-configuration performance, and CLAP B1 uses a same-family scorer.

values of 0.062–0.088, corresponding to relative decreases of 23–53%. B3 decreases from raw values of 0.161–0.185 to projected values of 0.090–0.112, corresponding to relative decreases of 31–48%. B2 decreases for four encoders, whereas Wav2CLIP B2 is 0.211 before and after projection at printed precision. Paired 95% bootstrap intervals over 1,010 clip-segments exclude zero for every B1 and B3 cell and for four of five B2 cells. The projected captions therefore show no compensating B2 increase on this evaluation pool.

Table 2 gives the budget comparison. At $k=15$, projected B1 ranges from -0.001 to 0.013 , whereas raw B1 ranges from 0.038 to 0.058 . Raw-token B1 exceeds projected-token B1 for every encoder at both token budgets. B3 at $k=15$ decreases from raw values of 0.070 – 0.098 to projected values of 0.021 – 0.037 , so source recall follows B1 at the smaller budget as well. B2 at $k=15$ also decreases for every encoder, from 0.170 – 0.178 to 0.149 – 0.153 .

Figure 3 juxtaposes R@1 on 1,006 evaluation clips, B1 on 1,010 clip-segments, and representation geometry on 3,568 disjoint paired clips. Table 3 reports the geometry and B1 values by encoder. For each raw and projected encoder state, we compute CKA with the CLIP visual embeddings [29]. We also measure top-10 nearest-neighbor overlap, orthogonal residual, effective rank [30], and mean off-diagonal cosine anisotropy [31].

In Fig. 3, every arrow in the three left panels of row 1 points upward and every arrow in the three right panels points downward. Row 2 shows the same directions for video-to-audio R@1 and for A2V R@10. Anisotropy decreases from 0.03 – 0.99 for raw states to 0.004 – 0.03 for projected states, and nearest-neighbor overlap with the visual graph rises for every encoder while remaining below 0.08 . Projection raises CKA with visual embeddings and visual nearest-

neighbor overlap for every encoder. Raw orthogonal residuals range from 0.61 to 0.69 , whereas projected residuals range from 0.25 to 0.32 . Nearest-neighbor overlap with the raw audio graph decreases from 1.00 for each raw state to 0.12 – 0.33 for projected states. At the encoder-state level, B1 decreases alongside the orthogonal residual and nearest-neighbor overlap with the raw audio graph for all five encoders. Effective rank increases and anisotropy decreases for every encoder, so the B1 decreases do not coincide with effective-rank collapse or increased anisotropy. These joint shifts in Fig. 3 are state-level associations and do not identify a causal mechanism. With ten encoder states and collinear geometric quantities, we do not fit multivariable coefficients.

A cross-backbone evaluation tests whether the main-backbone direction recurs elsewhere. The five frozen backbones are LLaVA-1.6-Mistral-7B [1, 2], LLaVA-1.5-7B [10], LLaVA-1.6-Vicuna-7B [1], LLaVA-1.5-13B [10], and LLaVA-1.6-34B [1]. We call the fixed 150-clip subset used for AudioCLIP and ImageBind on the two additional 7B backbones the *companion split*.

Across the two additional 7B backbones and five encoders, projected-token B1 is lower than raw-token B1 in eight of ten cells. Both exceptions occur on LLaVA-1.5-7B. They are Whisper, from 0.023 to 0.056 , and Wav2CLIP, whose projected B1 exceeds raw B1 by 0.003 . BLEU, ROUGE-1, and ROUGE-L favor raw tokens in ten of ten cells. ROUGE-2 favors raw tokens in nine of ten cells. On LLaVA-1.6-34B, raw tokens score higher than projected tokens in 30 of 30 available checks under partial coverage, comprising 25 text-reference checks and five caption-to-image CLIPScore checks. The five-encoder 34B extension was not evaluated with B1, B2, or B3, and its main-pool coverage is about 12–13% of clips, with complete

Table 1. Raw and projected tokens are compared at $k=150$. R@1 is reported in percent on the 1,006-clip evaluation pool. B1, B2, and B3 are computed over 1,010 clip-segments. CLAP and Whisper R@1 values are lower bounds for their original encoder configurations. Because B1 and the CLAP encoder use the same checkpoint family, interpret the CLAP B1 row with B3.

Encoder	R@1 (%) raw→projected	B1 raw→projected	B2 raw→projected	B3 raw→projected
AudioCLIP	0.00→1.19	.125→.088	.219→.204	.161→.111
Wav2CLIP	0.00→3.88	.113→.086	.211→.211	.163→.112
ImageBind	0.10→ 15.71	.132→.078	.226→.198	.169→.094
CLAP	0.10→1.99	.129→.071	.225→.199	.185→.107
Whisper	0.10→0.70	.131→.062	.223→.192	.174→.090

Table 2. Each entry gives the raw-to-projected change in B1 or B3 at one token budget. At $k=15$, projected-token B1 is at most 0.013, while raw-token B1 ranges from 0.038 to 0.058. Because B1 and the CLAP encoder use the same checkpoint family, interpret the CLAP B1 row with B3.

Encoder	B1 $k=15$	B1 $k=150$	B3 $k=15$	B3 $k=150$
AudioCLIP	.039→.007	.125→.088	.070→.029	.161→.111
Wav2CLIP	.038→.013	.113→.086	.071→.037	.163→.112
ImageBind	.058→-.001	.132→.078	.082→.023	.169→.094
CLAP	.057→.001	.129→.071	.098→.021	.185→.107
Whisper	.058→.004	.131→.062	.073→.023	.174→.090

Table 3. Geometry is computed on 3,568 paired clips, whereas B1 is computed on 1,010 clip-segments at $k=150$. Centered kernel alignment (CKA) is computed with CLIP visual embeddings. Orthogonal residual and effective rank follow Sec. 2. Interpret CLAP B1 with B3 because the encoder and scorer share a checkpoint family.

Encoder	Visual CKA	Orth. residual	Eff. rank	B1
AudioCLIP	.06→.19	.64→.32	11→47	.125→.088
Wav2CLIP	.17→.25	.61→.27	57→109	.113→.086
ImageBind	.41→.42	.61→.25	115→132	.132→.078
CLAP	.20→.26	.63→.26	60→105	.129→.071
Whisper	.08→.30	.69→.28	7→65	.131→.062

coverage of the 150-clip companion split. A separate 60-item CLAP smoke pair has grounded metrics but is not part of the 30-check tally.

LLaVA-1.5-13B exhibits a separate generation-instability result. In a 60-clip check, raw tokens produce empty or repetitive captions for 15 of 60 clips, or 25%, compared with 2 of 60 clips, or 3%, for projected tokens. Projected tokens have higher lexical scores than raw tokens for three of five encoder pairs, while B1 favors raw tokens in four of five pairs. On that backbone, raw and projected B1 are 0.014 and 0.002 for AudioCLIP, 0.081 and 0.059 for Wav2CLIP, 0.023 and -0.011 for ImageBind, and 0.072 and 0.046 for CLAP, whereas Whisper is the exception at 0.015 and 0.049. In the separate 60-item CLAP pair on LLaVA-1.6-34B, B1 is 0.187 for raw and 0.079 for projected tokens. Across the five frozen backbones from 7B to 34B, raw-over-projected generation scores predominate wherever raw-token generation is stable.

At sample sizes of 500, 1,000, 2,000, and 3,568 paired clips, every raw-to-projected geometry contrast retains its direction. The fifteen within-cell tests cover six CLAP, Whisper, and Wav2CLIP cells at 500 clips per cell. No association between per-sample geometry and caption quality survives Bonferroni correction. The geometry evidence therefore supports state-level associations only.

Table 4. Retrieval beyond R@1 on the 1,006-clip evaluation pool, in percent, raw→projected. A2V is audio-to-video and V2A is video-to-audio. CLAP and Whisper values are lower bounds for their original encoder configurations.

Encoder	A2V R@5	A2V R@10	V2A R@1	V2A R@10
AudioCLIP	0.00→5.07	0.99→7.55	0.00→1.19	0.99→9.24
Wav2CLIP	0.30→13.22	0.50→21.27	0.30→4.47	0.99→20.58
ImageBind	0.60→37.57	0.80→50.20	0.10→17.40	0.99→49.50
CLAP	0.30→7.16	0.70→11.93	0.10→2.19	1.09→13.32
Whisper	0.50→2.19	0.80→3.88	0.00→1.69	0.89→8.85

Vision-only references can favor projected captions on the companion split because a projected caption can closely reproduce the reference. For the Qwen2.5-Omni references [32], an automated audit finds an audio-related keyword in 99.2% of covered clips. Raw-token BLEU exceeds projected-token BLEU in nine of ten primary encoder-budget cells. The Wav2CLIP $k=150$ cell is the exception at a delta of -0.0006. B1, B2, and B3 use no references.

Across the five encoder pairs, projection raises R@1 while B1, B3, orthogonal residual, and nearest-neighbor overlap with the raw audio graph decrease. ImageBind has the largest R@1 change, from 0.10% to 15.71%. Whisper has the largest B1 change, from 0.131 to 0.062, corresponding to a 53% relative decrease. These paired encoder-level contrasts do not identify a causal mechanism.

5. CONCLUSION

Across five frozen encoders on one evaluation pool, projection into CLIP’s visual space increases audio-to-video R@1 and decreases B1 and B3. The retrieval gain holds at R@5 and R@10 and in the video-to-audio direction. Paired geometry contrasts associate lower B1 with lower orthogonal residual and lower nearest-neighbor overlap with the raw audio graph, while effective rank increases. These results do not identify a causal geometric mechanism, but they show that visual alignment and audio-native neighborhood structure capture different properties of the tested tokens. Across five frozen backbones from 7B to 34B, raw tokens generally receive higher grounded or reference-based generation scores than projected tokens wherever raw-token generation is stable. Within the tested configurations, the projected token is the better choice for retrieval and the raw token for grounded generation, subject to the backbone-stability caveat. The study is limited to five encoders, five backbones, and one evaluation pool. We do not evaluate joint raw-and-projected tokens or infer mechanisms from attention patterns.

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